

A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis

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Summary

Background Deep learning offers considerable promise for medical diagnostics. We aimed to evaluate the diagnostic accuracy of deep learning algorithms versus health-care professionals in classifying diseases using medical imaging.

Methods In this systematic review and meta-analysis, we searched Ovid-MEDLINE, Embase, Science Citation Index, and Conference Proceedings Citation Index for studies published from Jan 1, 2012, to June 6, 2019. Studies comparing the diagnostic performance of deep learning models and health-care professionals based on medical imaging, for any disease, were included. We excluded studies that used medical waveform data graphics material or investigated the accuracy of image segmentation rather than disease classification. We extracted binary diagnostic accuracy data and constructed contingency tables to derive the outcomes of interest: sensitivity and specificity. Studies undertaking an out-of-sample external validation were included in a meta-analysis, using a unified hierarchical model. This study is registered with PROSPERO, CRD42018091176.

Findings Our search identified 31 587 studies, of which 82 (describing 147 patient cohorts) were included. 69 studies provided enough data to construct contingency tables, enabling calculation of test accuracy, with sensitivity ranging from 9·7% to 100·0% (mean 79·1%, SD 0·2) and specificity ranging from 38·9% to 100·0% (mean 88·3%, SD 0·1). An out-of-sample external validation was done in 25 studies, of which 14 made the comparison between deep learning models and health-care professionals in the same sample. Comparison of the performance between health-care professionals in these 14 studies, when restricting the analysis to the contingency table for each study reporting the highest accuracy, found a pooled sensitivity of 87·0% (95% CI 83·0–90·2) for deep learning models and 86·4% (79·9–91·0) for health-care professionals, and a pooled specificity of 92·5% (95% CI 85·1–96·4) for deep learning models and 90·5% (80·6–95·7) for health-care professionals.

Interpretation Our review found the diagnostic performance of deep learning models to be equivalent to that of health-care professionals. However, a major finding of the review is that few studies presented externally validated results or compared the performance of deep learning models and health-care professionals using the same sample. Additionally, poor reporting is prevalent in deep learning studies, which limits reliable interpretation of the reported diagnostic accuracy. New reporting standards that address specific challenges of deep learning could improve future studies, enabling greater confidence in the results of future evaluations of this promising technology.

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Introduction

The first paper indexed in MEDLINE with the MeSH term “artificial intelligence” (AI) dates back to 1951, when Fletcher described a tortoise robot in the seminal paper “Matter with mind; a neurological research robot”.¹ Today, more than 16 000 peer-reviewed scientific papers are published in the AI field each year, with countless more in the lay press.² The application of AI has already started to transform daily life through applications such as photo captioning, speech recognition, natural language translation, robotics, and advances in self-driving cars.^{3–9}

Many people anticipate similar success in the health sphere, particularly in diagnostics, and some have suggested that AI applications will even replace whole medical disciplines or create new roles for doctors to fulfil, such as “information specialists”.^{10–12}

Medical imaging is one of the most valuable sources of diagnostic information but is dependent on human interpretation and subject to increasing resource challenges. The need for, and availability of, diagnostic images is rapidly exceeding the capacity of available specialists, particularly in low-income and middle-income

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Research in context

Evidence before this study

Deep learning is a form of artificial intelligence (AI) that offers considerable promise for improving the accuracy and speed of diagnosis through medical imaging. There is a strong public interest and market forces that are driving the rapid development of such diagnostic technologies. We searched Ovid-MEDLINE, Embase, Science Citation Index, and Conference Proceedings Citation Index for studies published from Jan 1, 2012, to June 6, 2019, that developed or validated a deep learning model for the diagnosis of any disease feature from medical imaging material and histopathology, with no language restrictions. We prespecified the cutoff of Jan 1, 2012, to reflect a recognised change in model performance with the development of deep learning approaches. We found that an increasing number of primary studies are reporting diagnostic accuracy of algorithms to be equivalent or superior when compared with humans; however, there are concerns around bias and generalisability. We found no other systematic reviews comparing performance of AI algorithms with health-care professionals for all diseases. We did find two disease-specific systematic reviews, but these mainly reported algorithm performance alone rather than comparing performance with health-care professionals.

Added value of this study

This review is the first to systematically compare the diagnostic accuracy of all deep learning models against

health-care professionals using medical imaging published to date. Only a small number of studies make direct comparisons between deep learning models and health-care professionals, and an even smaller number validate these findings in an out-of-sample external validation. Our exploratory meta-analysis of the small selection of studies validating algorithm and health-care professional performance using out-of-sample external validations found the diagnostic performance of deep learning models to be equivalent to health-care professionals. When comparing performance validated on internal versus external validation, we found that, as expected, internal validation overestimates diagnostic accuracy for both health-care professionals and deep learning algorithms. This finding highlights the need for out-of-sample external validation in all predictive models.

Implications of all the available evidence

Deep learning models achieve equivalent levels of diagnostic accuracy compared with health-care professionals. The methodology and reporting of studies evaluating deep learning models is variable and often incomplete. New international standards for study protocols and reporting that recognise specific challenges of deep learning are needed to ensure quality and interpretability of future studies.

countries.¹³ Automated diagnosis from medical imaging through AI, especially in the subfield of deep learning, might be able to address this problem.^{14,15} Reports of deep learning models matching or exceeding humans in diagnostic performance has generated considerable excitement, but this enthusiasm should not overrule the need for critical appraisal. Concerns raised in this field include whether some study designs are biased in favour of the new technology, whether the findings are generalisable, whether the study was performed in silico or in a clinical environment, and therefore to what degree the study results are applicable to the real-world setting. More than 30 AI algorithms have now been approved by the US Food and Drug Administration.¹⁶ In anticipation of AI diagnostic tools becoming implemented in clinical practice, it is timely to systematically review the body of evidence supporting AI-based diagnosis across the board.

In this systematic review, we have sought to critically appraise the current state of diagnostic performance by deep learning algorithms for medical imaging compared with health-care professionals, considering issues of study design, reporting, and clinical value to the real world, and we have conducted a meta-analysis to assess the diagnostic accuracy of deep learning algorithms compared with health-care professionals.

Methods

Search strategy and selection criteria

In this systematic review and meta-analysis, we searched for studies that developed or validated a deep learning model for the diagnosis of any disease feature from medical imaging material and histopathology, and additionally compared the accuracy of diagnoses made by algorithms versus health-care professionals. We searched Ovid-MEDLINE, Embase, Science Citation Index, and Conference Proceedings Citation Index for studies published from Jan 1, 2012, to June 6, 2019, with no language restrictions. The full search strategy for each database is available in the appendix (p 2). The cutoff of Jan 1, 2012, was prespecified on the basis of a recognised step-change in machine learning performance with the development of deep learning approaches. In 2012, for the first time, a deep learning model called AlexNet, enabled by advances in parallel computing architectures, made an important breakthrough at the ImageNet Large-Scale Visual Recognition Challenge.³ The search was first performed on and up to May 31, 2018, and an updated search was performed on June 6, 2019. Manual searches of bibliographies, citations, and related articles (PubMed function) of included studies were undertaken to identify any additional relevant articles that might have been missed by the searches.

See Online for appendix

Eligibility assessment was done by two reviewers who screened titles and abstracts of the search results independently, with non-consensus being resolved by a third reviewer. We did not place any limits on the target population, the disease outcome of interest, or the intended context for using the model. For the study reference standard to classify absence or presence of disease, we accepted standard-of-care diagnosis, expert opinion or consensus, and histopathology or laboratory testing. We excluded studies that used medical waveform data graphics material (ie, electroencephalography, electrocardiography, visual field data) or investigated the accuracy of image segmentation rather than disease classification.

Letters, preprints, scientific reports, and narrative reviews were included. Studies based on animals or non-human samples or that presented duplicate data were excluded.

This systematic review was done following the recommendations of the PRISMA statement.¹⁷ Methods of analysis and inclusion criteria were specified in advance. The research question was formulated according to previously published recommendations for systematic reviews of prediction models (CHARMS checklist).¹⁸

Data analysis

Two reviewers (XL, then one of LF, SKW, DJF, AK, AB, or TM) extracted data independently using a predefined data extraction sheet, cross-checked the data, and resolved disagreements by discussion or referral to a third reviewer (LMB or AKD). We contacted four authors for further information.^{19–22} One provided numerical data that had only been presented graphically in the published paper and one confirmed an error in their published contingency table. We did not formally assess the quality of the included studies.

Where possible, we extracted binary diagnostic accuracy data and constructed contingency tables at the reported thresholds. Contingency tables consisted of true-positive, false-positive, true-negative, and false-negative results, and were used to calculate sensitivity and specificity.

To estimate the accuracy of deep learning algorithms and health-care professionals, we did a meta-analysis of studies providing contingency tables from out-of-sample external validations (including geographical and temporally split data). If a study provided various contingency tables for the same or for different algorithms, we assumed these to be independent from each other. We accepted this assumption because we were interested in providing an overview of the results of various studies rather than providing precise point estimates. We used a unified hierarchical model that was developed for the meta-analysis of diagnostic accuracy studies and plotted summary receiver operating characteristic (ROC) curves for the accuracy of health-care professionals and deep

learning algorithms.²³ The hierarchical model involves statistical distributions at two different levels. At the lower level, it models the cell counts that form the contingency tables (true positive, true negative, false positive, and false negative) by using binomial distributions. This accounts for the within-study variability. At the higher level, it models the between-study variability (sometimes called heterogeneity) across studies. The hierarchical summary ROC figures provide estimates of average sensitivity and specificity across included studies with a 95% confidence region of the summary operating point and the 95% prediction region, which represents the confidence region for forecasts of sensitivity and specificity in a future study.

Owing to the broad nature of the review—ie, in considering any classification task using imaging for any disease—we were accepting of a large degree of between-study heterogeneity and thus it was not formally assessed.

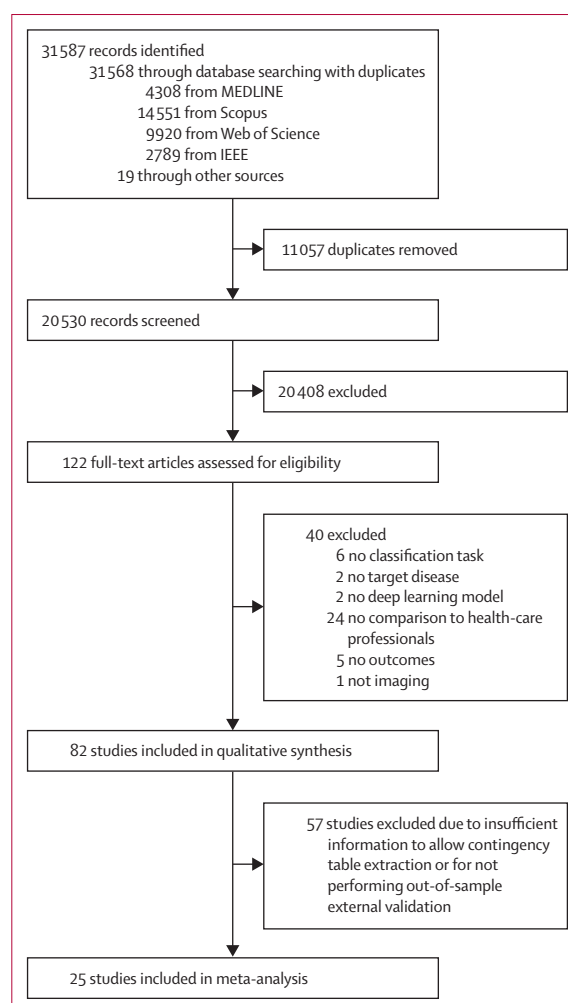


Figure 1: Study selection

IEEE=Institute of Electrical and Electronics Engineers.

	Subspecialty	Participants				
		Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data
Abbasi-Sureshjani et al (2018) ²⁴	Ophthalmology	NR	NR	NR (NR; 40–76)	51%	NR
Adams et al (2019) ²⁵	Trauma and orthopaedics	Emergency cases of surgically confirmed neck of femur fractures	Other radiological pathology present (excluding osteoporosis or osteoarthritis); metal-wear in fractured or unfractured hip	NR	NR	NR
Ardila et al (2019) ¹⁹	Lung cancer	Lung cancer screening patients	Unmatched scans to radiology reports; patients >1 year of follow-up	NR	NR	12 504
Ariji et al (2019) ²⁶	Oral cancer	Patients with contrast-enhanced CT and dissection of cervical lymph nodes	NR	Median 63 (NR; 33–95)	47%	NR
Ayed et al (2015) ²⁷	Breast cancer	NR	NR	NR (NR; 24–88)	100%	NR
Becker et al (2017) ²⁸	Breast cancer	Mammograms with biopsy proven malignant lesions	Surgery before first mammogram; metastatic malignancy involving breasts; cancer >2 years on external mammogram; in non-malignant cases, patients with <2 years of follow-up	57 (9; 32–85)	100%	2038
Becker et al (2018) ³⁰	Breast cancer	Mammograms with biopsy proven malignant lesions	Normal breast ultrasound or benign lesions, except if prior breast-conserving surgery was done; no radiological follow-up >2 years or histopathology proof	53 (15; 15–91)	100%	NR
Bien et al (2018) ²⁹	Trauma and orthopaedics	NR	NR	38 (NR; NR)	41%	1199
Brinker et al (2019) ³⁰	Dermatological cancer	NR	NR	NR	NR	NR
Brown et al (2018) ²¹	Ophthalmology	NR	Stage 4–5 retinopathy of prematurity	NR	NR	898
Burlina et al (2017) ³¹	Ophthalmology	NR	NR	NR	NR	NR
Burlina et al (2018) ³²	Ophthalmology	NR	NR	NR	NR	4152
Burlina et al (2018) ³³	Ophthalmology	NR	NR	NR	NR	4152
Byra et al (2019) ³⁴	Breast cancer	Masses with images in at least two ultrasound views	Inconclusive pathology; artifacts or known cancers	NR	NR	NR
Cao et al (2019) ³⁵	Urology	Patients undergoing robotic assisted laparoscopic prostatectomy with pre-operative MRI scans	Patients with prior radiotherapy or hormonal therapy	NR	NR	NR
Chee et al (2019) ³⁶	Trauma and orthopaedics	Patients aged ≥16 years with hip pain with osteonecrosis of the femoral head on MRI	>30 days between anteroposterior hip x-ray and hip MRI; history of hip operation with osseous abnormality in femoral head and neck; insufficient MRI and poor radiograph quality	48 (15; NR)	NR	NR
Choi et al (2019) ³⁷	Breast cancer	Patients aged ≥20 years with breast masses on ultrasound	Undiagnosed breast mass and low-quality images	Median 47 (NR; 42–54)	NR	NR
Choi et al (2018) ³⁸	Hepatology	Training set: pathologically confirmed cases External validation set: pathologically confirmed cases with no previous liver surgery and CT acquired within 5 months of examination	Tumour >5 cm; prior liver resection or transplant; anticancer treatment within 6 months of liver pathology; lymphoma or amyloidosis	Training dataset: 44 (15; 18–83) Test dataset 1: 48 (14; NR) Test dataset 2: 56 (10; NR) Test dataset 3: 53 (15; NR)	Training dataset: 28% Total test datasets: 43%	7461

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	Subspecialty	Participants				
		Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data
(Continued from previous page)						
Ciampi et al (2017) ³⁹	Respiratory disease	Baseline CT scans from the Multicentric Italian Lung Detection trial	Lesion diameter <4 mm	NR	NR	943
Codella et al (2017) ⁴⁰	Dermatological cancer	NR	NR	NR	NR	NR
Coudray et al (2018) ⁴¹	Lung cancer	NR	NR	NR	NR	NR
De Fauw et al (2018) ⁴²	Ophthalmology	All routine OCT images	Conditions with <10 cases	NR	Training dataset: 54% Test dataset: 55%	7621
Ding et al (2019) ⁴³	Neurology, psychiatry	Patients participating in the Alzheimer's Disease Neuroimaging Initiative clinical trial	Patients with no PET study ordered	Male: 76 (NR; 55–93) Female: 75 (NR; 55–96)	47%	899
Dunnmon et al (2019) ⁴⁴	Respiratory disease	NR	Images which are not anteroposterior or posteroanterior views	NR	NR	200 000
Ehteshami Bejnordi et al (2017) ⁴⁵	Breast cancer	Patients having breast cancer surgery	Isolated tumour cells in a sentinel lymph node	NR	100%	NR
Esteva et al (2017) ⁴⁶	Dermatological cancer	NR	NR	NR	NR	NR
Fujioka et al (2019) ⁴⁷	Breast cancer	Breast ultrasound of benign or malignant masses confirmed by pathology; patients with minimum 2-year follow-up	Patients on hormonal therapy or chemotherapy; patients aged <20 years	Training dataset: 55 (13; NR) Test dataset: 57 (15; NR)	NR	237
Fujisawa et al (2019) ⁴⁸	Dermatological cancer	NR	NR	NR	NR	1842
Gómez-Valverde et al (2019) ⁴⁹	Ophthalmology	Aged 55–86 years in glaucoma detection campaign	Poor-quality images	NR	NR	NR
Grewal et al (2018) ⁵⁰	Trauma and orthopaedics	NR	NR	NR	NR	NR
Haenssle et al (2018) ⁵¹	Dermatological cancer	NR	NR	NR	NR	NR
Hamm et al (2019) ⁵²	Liver cancer	Untreated liver lesions, or treated lesions that showed progression, or recurrence post 1 year local or regional therapy	Atypical imaging features; patients aged <18 years	57 (14; NR)	48%	296
Han et al (2018) ⁵³	Dermatological cancer	All images from datasets	For the Asan dataset, postoperative images were excluded	Asan 1: 47 (23; NR) Asan 2: 41 (21; NR) Atlas: NR MED-NODE: NR Hallym: 68 (13; NR) Edinburgh: NR	Asan 1: 55% Asan 2: 57% Atlas: NR MED-NODE: NR Hallym: 52% Edinburgh: NR	NR
Han et al (2018) ⁵⁴	Dermatological cancer	For Inje, Hallym, and Seoul datasets: onychomycosis: positive potassium, oxygen, and hydrogen test or fungus culture result; or successful treatment with antifungal drugs; nail dystrophy: negative potassium, oxygen, and hydrogen test or culture result; unresponsiveness to antifungal medication; or responsiveness to a triamcinolone intralesional injection	Inadequate images and images of uncertain diagnosis	Asan 1: 41 (22; NR) Asan 2: 46 (20; NR) Inje 1: 48 (23; NR) Inje 2: 54 (20; NR) Hallym: 39 (15; NR) Seoul: 51 (20; NR)	Asan 1: 55% Asan 2: 59% Inje 1: 56% Inje 2: 48% Hallym: 47% Seoul: 54%	NR
Hwang et al (2018) ⁵⁵	Respiratory disease	Active pulmonary tuberculosis ≤1 month from treatment initiation	Non-parenchymal tuberculosis and non-tuberculosis chest x-rays	51 (16; NR)	82%	NR
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	Subspecialty	Participants				
		Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data
(Continued from previous page)						
Hwang et al (2019) ⁵⁶	Ophthalmology	Age-related macular degeneration cases presenting to the hospital	Low-resolution images or improper format	NR	NR	NR
Hwang et al (2019) ⁵⁷	Respiratory disease	Cases of clinically or microbiologically confirmed pneumonia or clinically reported pneumothorax; cases of pulmonary tuberculosis (where a chest x-ray was completed within 2 weeks of treatment initiation)	Chest x-rays >3 lesions for lung cancer; pneumothorax chest x-rays with drainage catheter or subcutaneous emphysema	Training dataset: 51 (16; NR) normal images; 62 (15; NR) for abnormal images	Training dataset: 55% Test dataset: 38%	NR
Kermany et al (2018) ⁵⁸	Ophthalmology, respiratory disease	OCT: routine OCTs from local databases for choroidal neovascularisation, DMO, drusen, and normal images Chest x-rays: retrospective cohort of 1–5 year olds	OCT: none Chest x-rays: NR	Choroidal neovascularisation 1: 83 (NR; 58–97) DMO 2: 57 (NR; 20–90) Drusen: 82 (NR; 40–95) Normal: 60 (NR; 21–68) X-ray: NR	Choroidal neovascularisation 1: 46% DMO 2: 62% Drusen: 56% Normal: 41% X-ray: NR	OCT: 4686 Chest x-ray: 5856
Kim et al (2012) ⁵⁹	Breast cancer	Patients with solid mass on ultrasound	Breast Imaging Reporting and Data System: 0, 1, and 6	44 (NR, 22–70)	NR	70
Kim et al (2018) ⁶⁰	Trauma and orthopaedics	Tuberculous or pyogenic spondylitis	Unconfirmed diagnosis; no pre-diagnostic MRI; early postoperative infection and cervical infectious spondylitis	Tuberculous spondylitis: 59 (NR; 38–71) Pyogenic spondylitis: 64 (NR; 56–72)	Tuberculous spondylitis: 49% Pyogenic spondylitis: 40%	NR
Kim et al (2019) ⁶¹	Maxillofacial surgery	Age >16 years with suspected maxillary sinusitis with a Waters’ view plain film radiographs	History of sinus surgery, fracture, or certain tumours involving the maxillary sinus	Training dataset: 47 (20; NR) Test dataset: internal validation: 54 (21; NR); external validation: temporal 49 (20; NR), geographical: 53 (18; NR)	Training dataset: 54% Test dataset: internal validation: 56%; external validation: temporal 47%, geographical 54%	NR
Kise et al (2019) ⁶²	Rheumatology	Sjogren’s syndrome	NR	Sjogren’s syndrome: 67 (NR; NR) Control: 66 (NR; NR)	Sjogren’s syndrome: 4% Control: 97%	40
Ko et al (2019) ⁶³	Thyroid cancer	Ultrasound and subsequent thyroidectomy, nodules 1–2 cm with correlating pathology results	NR	Training dataset: 48 (13; 12–79) Test dataset: 50 (12; NR)	Training dataset: 82% Test dataset: 85%	NR
Kumagai et al (2019) ⁶⁴	Oesophageal cancer	NR	NR	NR	NR	240
Lee et al (2019) ⁶⁵	Trauma and orthopaedics	Training and test data: non-contrast head CT with or without acute ICH Prospective test data: non-contrast head CT in 4 months from the local hospital’s emergency department	History of brain surgery, skull fracture, intracranial tumour, intracranial device, cerebral infarct, or non-acute ICH	NR	NR	NR
Li C et al (2018) ⁶⁶	Nasopharyngeal cancer	Nasopharyngeal endoscopic images for screening	Blurred images or images with incomplete exposure	Training dataset: 46 (13; NR) Test dataset: 46 (13; NR) Prospective test dataset: 48 (13; NR)	Training dataset: 30% Test dataset: 32% Prospective test dataset: 34%	5557

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Subspecialty	Participants					
	Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data	
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Li X et al (2019) ⁶⁷	Thyroid cancer	Patients aged ≥18 years with thyroid cancer; patients with pathological examination and negative controls	Patients with thyroid cancer with differing pathological report	Training dataset: median 44 (NR; 36–54) Test dataset: internal validation: median 47 (NR; 24–41); external validation 1: median 51 (NR; 45–59); external validation 2: median 50 (NR; 41–59)	Training dataset: 75% Test dataset: internal validation: 77%; external validation 1: 78%; external validation 2: 80%	42 952
Lin et al (2014) ⁶⁸	Breast cancer	Solid mass on ultrasound	NR	52 (NR; NR)	100%	NR
Lindsey et al (2018) ⁶⁹	Trauma and orthopaedics	NR	NR	NR	NR	NR
Long et al (2017) ⁷⁰	Ophthalmology	Routine examinations done as part of the Childhood Cataract Program of the Chinese Ministry of Health, and search engine images matching the key words “congenital”, “infant”, “paediatric cataract”, and “normal eye”	NR	NR	NR	NR
Lu W et al (2018) ⁷¹	Ophthalmology	Image containing only one of the four abnormalities (serous macular detachment, cystoid macular oedema, macular hole, and epiretinal membrane)	Images with other abnormalities than the four included or co-existence of two abnormalities	NR	NR	NR
Matsuba et al (2019) ⁷²	Ophthalmology	Men aged >70 years and women aged >77 years	Unclear images due to vitreous haemorrhage, astrocytosis, or strong cataracts; previous retinal photocoagulation and other complicating fundus disease as determined by retinal specialists	Control: 77 (5; NR) Wet age-related macular degeneration: 76 (82; NR)	Control: 28% Wet age-related macular degeneration: 26%	NR
Nakagawa et al (2019) ⁷³	Oesophageal cancer	Patients with superficial oesophageal squamous cell carcinoma with pathologic proof of cancer invasion depth	Severe oesophagitis; oesophagus chemotherapy or radiation history; lesions adjacent to ulcer or ulcer scar	Median 69 (NR; 44–90)	21%	NR
Nam et al (2019) ⁷⁴	Lung cancer	Training: malignant lung nodules chest x-rays proven by histopathology External validation: chest x-rays with referential normal CTs performed within 1 month	Nodules ≤5 mm on CT, chest x-rays showing ≥3 nodules, lung consolidation, or pleural effusion obscuring view	Female: 52 (NR) Male: 53 (NR)	Normal: 45% Abnormal: 42%	NR
Olczak et al (2017) ⁷⁵	Trauma and orthopaedics	NR	NR	NR	NR	NR
Peng et al (2019) ⁷⁶	Ophthalmology	NR	NR	NR	NR	4099
Poedjastoeti et al (2018) ⁷⁷	Oral and maxillofacial cancer	Panoramic x-rays of ameloblastomas and keratocystic odontogenic tumours with known biopsy results	NR	NR	NR	NR
Rajpurkar et al (2018) ⁷⁸	Respiratory disease	NR	NR	NR	NR	NR
Raumviboonsuk et al (2019) ⁷⁹	Ophthalmology	NR	Pathologies precluding classification of target condition, or presence of other retinal vascular disease	61 (11; NR)	67%	NR
Sayres et al (2019) ⁸⁰	Ophthalmology	NR	NR	NR	NR	NR
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	Subspecialty	Participants				
		Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data
(Continued from previous page)						
Schlegl et al (2018) ³²	Ophthalmology	Random sample of age-related macular degeneration, DMO, and retinal vein occlusion cases	No clear consensus or poor image quality	NR	NR	NR
Shibutani et al (2019) ⁸¹	Cardiology	Myocardial perfusion SPECT within 45 days of coronary angiography	NR	72 (9; 50–89)	19%	NR
Shichijo et al (2017) ⁸²	Gastroenterology	A primary care referral for OGD for epigastric symptoms, barium meal results, abnormal pepsinogen levels, previous gastroduodenal disease, or screening for gastric cancer	<i>Helicobacter pylori</i> eradication; presence or history of gastric cancer, ulcer, or submucosal tumour; unclear images	Training dataset: 53 (13; NR) Test dataset: 50 (11; NR)	Training dataset: 55% Test dataset: 57%	735
Singh et al (2018) ⁸³	Respiratory disease	Randomly selected chest x-rays from the database	Lateral radiographs; oblique views; patients with total pneumonectomy; patients with a metal prosthesis	NR	NR	NR
Song et al (2019) ⁸⁴	Thyroid cancer	Patients aged >18 years with total or nearly total thyroidectomy or lobectomy, with complete preoperative thyroid ultrasound images with surgical pathology examination	Failure to meet American Thyroid Association criteria for lesions or nodules	Training dataset: NR (NR; NR) Test dataset: 57 (16; NR)	Training dataset: NR Test dataset: 90%	NR
Stoffel et al (2018) ⁸⁵	Breast cancer	Ultrasound scan and histologically confirmed phyllodes tumour and fibroadenoma	NR	34 (NR; NR)	NR	NR
Streba et al (2012) ⁸⁶	Hepatological cancer	Patients with suspected liver masses (with hepatocellular carcinoma, hypervascular and hypovascular liver metastases, hepatic haemangiomas, or focal fatty changes) who underwent contrast-enhanced ultrasound	NR	58 (NR; 29–89)	43%	NR
Sun et al (2014) ⁸⁷	Cardiology	Patients with paroxysmal atrial fibrillation or persistent atrial fibrillation	NR	60 (11; 29–81)	45%	NR
Tschandl et al (2019) ⁸⁸	Dermatological cancer	Lesions that had lack of pigment, availability of at least one clinical close-up image or one dermatoscopic image, and availability of an unequivocal histopathologic report	Mucosal or missing or poor image cases; equivocal histopathologic reports; cases with <10 examples in the training set category	NR	NR	NR
Urakawa et al (2019) ⁸⁹	Trauma and orthopaedics	All consecutive patients with intertrochanteric hip fractures, and anterior x-ray with compression hip screws	Pseudarthrosis after femoral neck fracture or x-rays showing artificial objects in situ	85 (NR; 29–104)	84%	NR
van Grinsven et al (2016) ⁹⁰	Ophthalmology	NR	NR	NR	NR	NR
Walsh et al (2018) ⁹¹	Respiratory disease	High-resolution CT showing diffuse fibrotic lung disease confirmed by at least two thoracic radiologists	Contrast-enhanced CT	NR	NR	NR
Wang et al (2017) ⁹²	Lung cancer	NR	PET/CT scan in lobectomy patients with systematic hilar and mediastinal lymph node dissection	61 (NR; 38–81)	46%	NR

(Table 1 continues on next page)

(Table 1 continues on next page)

	Subspecialty	Participants				
		Inclusion criteria	Exclusion criteria	Mean age (SD; range), years	Percentage of female participants	Number of participants represented by the training data
(Continued from previous page)						
Wang et al (2018) ⁹³	Lung cancer	Solitary pulmonary nodule, histologically confirmed pre-invasive lesions and invasive adenocarcinomas	Previous chemotherapy or radiotherapy that can cause texture changes; incomplete CT; patients with ≥2 lesions resected	56 (10.6; NR)	81%	NR
Wang et al (2019) ⁹⁴	Thyroid cancer	Ultrasound examination with subsequent histological diagnosis	NR	46 (NR; 20–71)	NR	NR
Wright et al (2014) ⁹⁵	Nephrology	NR	Equivocal reports; artefacts; bladder inclusion and residual uptake in the ureters; horseshoe kidney	9 (NR; 0–80)	70%	257
Wu et al (2019) ⁹⁶	Gastric cancer	Patients undergoing OGD	Age <18 years; residual stomach content	NR	NR	NR
Ye et al (2019) ⁹⁷	Trauma and orthopaedics	Patients with ICH	Missing information or serious imaging artefact	Non-ICH :42 (15; 2–82) ICH: 54 (17; 1–98)	Non-ICH: 55% ICH: 35%	NR
Yu et al (2018) ⁹⁸	Dermatological cancer	Benign nevi or acral melanoma with histological diagnosis and dermatoscopic images	NR	NR	NR	NR
Zhang C et al (2019) ⁹⁹	Lung cancer	CT scans from lung cancer screening	Images with no ground truth labels available	60 (11; NR)	Training: 44%	NR
Zhang Y et al (2019) ¹⁰⁰	Paediatrics, ophthalmology	NR	Blurry, very dark or bright, or non-fundus images were excluded	NR	56%	17 801
Zhao et al (2018) ¹⁰¹	Lung cancer	Thin-slice chest CT scan before surgical treatment; nodule diameter ≤10 mm on CT; no treatment before surgical treatment	NR	54 (12; 16–82)	NR	NR
NR=not reported. OCT=optical coherence tomography. DMO=diabetic macular oedema. ICH=intracranial haemorrhage. SPECT=single-photon-emission CT. OGD=oesophagogastrroduodenoscopy.						
Table 1: Participant demographics for the 82 included studies						

To estimate the accuracy of deep learning algorithms compared with health-care professionals, we did a subanalysis for studies providing contingency tables for both health-care professional and deep learning algorithm performance tested using the same out-of-sample external validation datasets. Additionally, to address the possibility of dependency between different classification tasks done by the same deep learning algorithm or health-care professional within a study, we did a further analysis on the same studies selecting the single contingency table reporting the highest accuracy for each (calculated as proportion of correct classifications).

As an exploratory analysis, we also pooled performances of health-care professionals and deep learning algorithms derived from internally validated test samples. As with the externally validated results, we selected a single contingency table for each study reporting the highest accuracy for health-care professionals and deep learning algorithms. The purpose of this analysis was to explore whether diagnostic accuracy is overestimated in internal validation alone.

Analysis was done using the Stata 14.2 statistics software package. This study is registered with PROSPERO, CRD42018091176.

Role of the funding source

There was no funding source for this study. The lead authors (XL, LF) and senior author (AKD) had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

Our search identified 31 587 records, of which 20 530 were screened (figure 1). 122 full-text articles were assessed for eligibility and 82 studies were included in the systematic review.^{19–22,24–101} These studies described 147 patient cohorts and considered ophthalmic disease (18 studies), breast cancer (ten studies), trauma and orthopaedics (ten studies), dermatological cancer (nine studies), lung cancer (seven studies), respiratory disease (eight studies), gastroenterological or hepatological cancers (five studies), thyroid cancer (four studies), gastroenterology and hepatology (two studies), cardiology (two studies),

	Target condition	Reference standard	Same method for assessing reference standard across samples	Type of internal validation	External validation
Abbasi-Sureshjani et al (2018) ²⁴	Diabetes	Laboratory testing	Yes	Random split sample validation	No
Adams et al (2019) ²⁵	Hip fracture	Surgical confirmation	Yes	Random split sample validation	No
Ardila et al (2019) ²⁹	Lung cancer	Histology; follow-up	No	NR	Yes
Ariji et al (2019) ²⁶	Lymph node metastasis	Histology	Yes	Resampling method	No
Ayed et al (2015) ²⁷	Breast tumour	Histology	Yes	Random split sample validation	No
Becker et al (2017) ²⁸	Breast tumour	Histology; follow-up	No	Study 1: NA Study 2: temporal split-sample validation	Yes
Becker et al (2018) ³⁰	Breast tumour	Histology; follow-up	No	Random split sample validation	No
Bien et al (2018) ²⁹	Knee injuries	Expert consensus	Internal validation dataset: yes External validation dataset: NR	Stratified random sampling	No
Brinker et al (2019) ³⁰	Melanoma	Histology	Yes	Random split sample validation	Yes
Brown et al (2018) ²¹	Retinopathy	Expert consensus	Yes	Resampling method	Yes
Burlina et al (2017) ³¹	Age-related macular degeneration	Expert consensus	Yes	Resampling method	No
Burlina et al (2018) ³²	Age-related macular degeneration	Reading centre grader	Yes	NR	No
Burlina et al (2018) ³³	Age-related macular degeneration	Reading centre grader	Yes	NR	No
Byra et al (2019) ³⁴	Breast tumour	Histology; follow-up	No	Resampling method	Yes
Cao et al (2019) ³⁵	Prostate cancer	Histology; clinical care notes or imaging reports	Yes	Resampling method	No
Chee et al (2019) ³⁶	Femoral head osteonecrosis	Clinical care notes or imaging reports	Yes	NR	Yes
Choi et al (2019) ³⁷	Breast tumour	Histology; follow-up	No	NA	Yes
Choi et al (2018) ³⁸	Liver fibrosis	Histology	Yes	Resampling method	Yes
Ciampi et al (2017) ³⁹	Lung cancer	Expert consensus	Yes	Random split sample validation	Yes
Codella et al (2017) ⁴⁰	Melanoma	Histology	No	Random split sample validation	No
Coudray et al (2018) ⁴¹	Lung cancer	Histology	Yes	NR	Yes
De Fauw et al (2018) ⁴²	Retinal disease	Follow-up	Yes	Random split sample validation	No
Ding et al (2019) ⁴³	Alzheimer's disease	Follow-up	No	NR	Yes
Dunnmon et al (2019) ⁴⁴	Lung conditions	Expert consensus	Yes	Resampling method	No
Ehteshami Bejnordi et al (2017) ⁴⁵	Lymph node metastases	Histology	No	Random split sample validation	Yes
Esteva et al (2017) ⁴⁶	Dermatological cancer	Histology	No	Resampling method	No
Fujioka et al (2019) ⁴⁷	Breast tumour	Histology; follow-up	No	NR	No
Fujisawa et al (2019) ⁴⁸	Dermatological cancer	Histology	No	Resampling method	No
Gómez-Valverde et al (2019) ⁴⁹	Glaucoma	Expert consensus	Yes	Resampling method	No
Grewal et al (2018) ⁵⁰	Brain haemorrhage	Expert consensus	Yes	NR	No
Haenssle et al (2018) ⁵¹	Melanoma	Histology; follow-up	No	NR	No
Hamm et al (2019) ⁵²	Liver tumour	Clinical care notes or imaging reports	Yes	Resampling method	No
Han et al (2018) ⁵³	Onychomycosis	Histology; expert opinion on photography	No	Random split sample validation	Yes
Han et al (2018) ⁵⁴	Skin disease	Histology; follow-up	No	Random split sample validation	Yes
Hwang et al (2018) ⁵⁵	Pulmonary tuberculosis	Laboratory testing; expert opinion	Yes	NR	Yes

(Table 2 continues on next page)

	Target condition	Reference standard	Same method for assessing reference standard across samples	Type of internal validation	External validation
(Continued from previous page)					
Hwang et al (2019) ⁵⁶	Age-related macular degeneration	Expert consensus	Yes	Random split sample validation	Yes
Hwang et al (2019) ⁵⁷	Lung conditions	Expert consensus	Yes	Random split sample validation	No
Kermany et al (2018) ⁵⁸	Retinal diseases	OCT: consensus involving experts and non-experts X-ray: expert consensus	No	Random split sample validation	No
Kim et al (2012) ⁵⁹	Breast cancer	Histology	Yes	Random split sample validation	No
Kim et al (2018) ⁶⁰	Maxillary sinusitis	Histology; laboratory testing	Yes	Resampling method	No
Kim et al (2019) ⁶¹	Spondylitis	Expert consensus; another imaging modality	Yes	NR	Yes
Kise et al (2019) ⁶²	Sjogren's syndrome	Expert consensus	Yes	NR	No
Ko et al (2019) ⁶³	Thyroid cancer	Histology	Yes	Resampling method	No
Kumagai et al (2019) ⁶⁴	Oesophageal cancer	Histology	Yes	NR	No
Lee et al (2019) ⁶⁵	Intracranial haemorrhage	Expert consensus	Yes	Random split sample validation	Yes
Li C et al (2018) ⁶⁶	Nasopharyngeal malignancy	Histology	Yes	Random split sample validation	Yes
Li X et al (2019) ⁶⁷	Thyroid cancer	Histology	Yes	NR	Yes
Lin et al (2014) ⁶⁸	Breast tumour	Histology	Yes	NR	No
Lindsey et al (2018) ⁶⁹	Trauma and orthopaedics	Expert consensus	Yes	NR	Yes
Long et al (2017) ⁷⁰	Ophthalmology	Expert consensus	Yes	Resampling method	Yes
Lu W et al (2018) ⁷¹	Macular pathology	Expert consensus	Yes	Resampling method	No
Matsuba et al (2019) ⁷²	Age-related macular degeneration	Expert consensus	Yes	NR	No
Nakagawa et al (2019) ⁷³	Oesophageal cancer	Histology	Yes	NR	Yes
Nam et al (2019) ⁷⁴	Lung cancer	Expert consensus; another imaging modality; clinical notes	No	Random split sample validation	Yes
Olczak et al (2017) ⁷⁵	Fractures	Clinical care notes or imaging reports	Yes	Random split sample validation	No
Peng et al (2019) ⁷⁶	Age-related macular degeneration	Reading centre grader	Yes	NR	No
Poedjiastoeti et al (2018) ⁷⁷	Odontogenic tumours of the jaw	Histology	Yes	NR	No
Rajpurkar et al (2018) ⁷⁸	Lung conditions	Expert consensus	Yes	NR	No
Raumviboonsuk et al (2019) ⁷⁹	Diabetic retinopathy	Expert consensus	Yes	NR	Yes
Sayres et al (2019) ⁸⁰	Diabetic retinopathy	Expert consensus	Yes	NR	No
Schlegl et al (2018) ⁷²	Macular diseases	Expert consensus	Yes	Resampling method	No
Shibutani et al (2019) ⁸¹	Myocardial stress defect	Expert consensus	Yes	NR	Yes
Shichijo et al (2017) ⁸²	<i>Helicobacter pylori</i> gastritis	Standard-of-care diagnosis based on laboratory testing	No	Random split sample validation	No
Singh et al (2018) ⁸³	Lung conditions	Clinical care notes or imaging reports; existing labels in open-access data library	No	NR	No
Song et al (2019) ⁸⁴	Thyroid cancer	Histology	Yes	Resampling method	No
Stoffel et al (2018) ⁸⁵	Breast tumours	Histology	Yes	Random split sample validation	No
Streba et al (2012) ⁸⁶	Liver tumours	Another imaging modality; histology; follow-up	No	Resampling method	No

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	Target condition	Reference standard	Same method for assessing reference standard across samples	Type of internal validation	External validation
(Continued from previous page)					
Sun et al (2014) ⁸⁷	Atrial thrombi	Surgical confirmation; another imaging modality; clinical care notes or imaging reports	No	Random split sample validation	No
Tschandl et al (2019) ⁸⁸	Dermatological cancer	Histology	Yes	NR	Yes
Urakawa et al (2019) ⁸⁹	Hip fractures	Clinical care notes or imaging reports	Yes	Random split sample validation	No
van Grinsven et al (2016) ⁹⁰	Retinal haemorrhage	Single expert	Yes	Random split sample validation	Yes
Walsh et al (2018) ⁹¹	Lung fibrosis	Expert consensus	Yes	NR	Yes
Wang et al (2017) ⁹²	Lymph node metastasis	Expert consensus	Yes	Resampling method	No
Wang et al (2018) ⁹³	Lung cancer	Histology	Yes	Random split sample validation	No
Wang et al (2019) ⁹⁴	Malignant thyroid nodule	Histology	Yes	NR	No
Wright et al (2014) ⁹⁵	Renal tissue function	Clinical care notes or imaging reports	Yes	Random split sample validation	No
Wu et al (2019) ⁹⁶	Gastric cancer	Histology	Yes	Resampling method	No
Ye et al (2019) ⁹⁷	Intracranial haemorrhage	Expert consensus	Yes	Random split sample validation	No
Yu et al (2018) ⁹⁸	Melanoma	Histology	Yes	Resampling method	No
Zhang C et al (2019) ⁹⁹	Lung cancer	Expert consensus	Yes	Resampling method	Yes
Zhang Y et al (2019) ¹⁰⁰	Retinopathy	Expert consensus	Yes	Random split sample validation	No
Zhao et al (2018) ¹⁰¹	Lung cancer	Histology	Yes	NR	No
Blinded assessment of reference standard was not reported in any of the studies. NR=not reported. OCT=optical coherence tomography. DMSA=2,3-dimercapto-succinic acid.					
Table 2: Model training and validation for the 82 included studies					

oral cancer (two studies), nephrology (one study), neurology (one study), maxillofacial surgery (one study), rheumatology (one study), nasopharyngeal cancer (one study), and urological disease (one study; table 1). One study included two different target conditions.⁵⁸ Study characteristics are summarised in the tables (tables 1, 2, 3).

72 studies used retrospectively collected data and ten used prospectively collected data (table 3). 25 studies used data from open-access repositories. No studies reported a prespecified sample size calculation. 26 studies reported that low-quality images were excluded, 18 did not exclude low-quality images, and 38 did not report this. Four studies^{19,42,51,80} also tested the scenario where health-care professionals are given additional clinical information alongside the image, and one study¹⁹ tested single image versus the addition of historical images for both health-care professionals and the deep learning algorithm. Four studies also considered diagnostic performance in an algorithm-plus-clinician scenario.^{69,74,85,87}

Reference standards were wide ranging in line with variation of the target condition and the modality of imaging being used, with some studies adopting multiple methods (table 2). 37 studies used histopathology; 28 studies used varying models of expert consensus; one

study relied on single expert consensus; nine studies used clinical follow-up; two studies used surgical confirmation; three studies used reading centre labels (such as when clinical trial data were used); eight studies used existing clinical care notes or imaging reports or existing labels associated with open data sources. Four studies used another imaging modality to confirm the diagnosis and four studies used laboratory testing.

69 studies provided sufficient information to enable calculation of contingency tables and calculation of test performance parameters, with a total of 595 tables across these studies. Within this group, sensitivity for deep learning models ranged from 9.7% to 100.0% (mean 79.1%, SD 0.2) and specificity ranged from 38.9% to 100.0% (mean 88.3%, SD 0.1).

Of the 69 studies, 25 studies did an out-of-sample external validation and were therefore included in a meta-analysis.^{21,28,30,34,36–39,43,53–56,61,65–67,70,73,74,79,81,90,91,99} In line with the aims of this review, all eligible studies were included regardless of the target condition. The meta-analysis therefore included diagnostic classifications in multiple specialty areas, including ophthalmology (six studies), breast cancer (three studies), lung cancer (two studies), dermatological cancer (three studies), trauma and orthopaedics (two studies), respiratory disease (two studies),

Indicator definition			Algorithm		Data source		Open-access data
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/tuning	
Abbasi-Sureshjani et al (2018) ³⁴	NR	NR	ResNet	CNN; Residual Network	No	7931/NR	Retrospective cohort; secondary analysis of a subset of the Maastricht study—a population-based cohort (collected in the southern part of the Netherlands), enriched with patients with diabetes
Adams et al (2019) ³⁵	NR	No	AlexNet	CNN; AlexNet	Yes	512/NR	Retrospective cohort; data from the Royal Melbourne Hospital (Melbourne, VIC, Australia) radiographic archive
Ardila et al (2019) ³⁶	No	Yes	Mask RCNN; RetinaNet; Inception V1	CNN; Inception	Yes	10 396/2228	Retrospective clinical trial data from the National Lung Cancer Screening Trial
Ariji et al (2019) ³⁶	NR	No	CNN	CNN	No	NR/NR	Retrospective cohort; data from the Aichi-Gakuin University School of Dentistry (Nagoya, Japan)
Ayed et al (2015) ³⁷	NR	NR	ANN	ANN	No	200/NR	Retrospective cohort; secondary analysis of a subset of the Farabi Digital Database for Screening Mammography collected at the radiology centre El Farabi (Tunisia)
Becker et al (2017) ³⁸	No	Yes	VIDI Suite Version 20	ANN	No	Study 1: 95/48 Study 2: 513/257	Study 1, cohort A: retrospective cohort; data collected at the University Hospital Zurich (Zurich, Switzerland) Study 1, cohort B: retrospective cohort; secondary analysis of the Breast Cancer Digital Repository Study 2: retrospective cohort; data collected at the University Hospital Zurich (Zurich, Switzerland)
Becker et al (2018) ³⁹	No	Yes	VIDI Suite Version 20	ANN	No	445/192	Retrospective cohort; data collected at the University Hospital Zurich (Zurich, Switzerland)
Bien et al (2018) ³⁹	No	Yes	MRNet	CNN	Yes	1130/120	Retrospective cohort; data from the Stanford University Medical Center (CA, USA)
Brinker et al (2019) ³⁹	NR	No	ResNet-50	CNN; Residual Network	Yes	12 378/1359	Retrospective cohort; data collected at multiple institutions for a research challenge (International Skin Image Collaboration)
Brown et al (2018) ⁴¹	Yes	No	CNN	CNN	Yes	4409/1102	Retrospective cohort; data collected at multiple hospitals across North America
Burlina et al (2017) ³¹	NR	No	AlexNet; OverFeat	CNN; AlexNet	Yes	5664/NR	Retrospective cohort; secondary analysis of a subset from the Age-related Eye Disease Study trial
Burlina et al (2018) ³⁷	NR	No	ResNet	CNN; Residual Network	Yes	59 313/1348	Retrospective cohort; secondary analysis of a subset from the Age-related Eye Disease Study trial
Burlina et al (2018) ³³	Yes	No	ResNet-50	CNN; Residual Network	Yes	59 313/1348	Retrospective cohort; secondary analysis of a subset from the Age-related Eye Disease Study trial

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Indicator definition			Algorithm		Data source		Open-access data			
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/tuning		Source of data	Data range	
(Continued from previous page)										
Byra et al (2019) ³⁴	No	No	VGG-19	CNN; VGG	Yes	582/150	Three data cohorts: Cohort 1: retrospective cohort; secondary analysis from the Moores Cancer Center, University of California (San Diego, CA, USA) Cohort 2: retrospective cohort; secondary analysis of the UDIAT Diagnostic Centre of the Parc Tauli Corporation (Sabadell, Spain) Cohort 3: Retrospective cohort; data collected from the Institute of Oncology (Warsaw, Poland)	Cohort 1: NR Cohort 2: 2012 Cohort 3: 2013–15	Cohort 1: no Cohort 2: yes Cohort 3: yes	
Cao et al (2019) ³⁵	MRI	NR	No	FocalNet	CNN	Yes	NR/NR	Retrospective cohort	NR	No
Chee et al (2019) ³⁶	MRI	Yes	Yes	ResNet	CNN; Residual Network	Yes	1346/148	Retrospective cohort; secondary analysis of data collected at Seoul National University Hospital and Seoul National University Bundang Hospital (Seoul, Korea)	2003–17	No
Choi et al (2019) ³⁷	Ultrasound	Yes	No	GoogLeNet	CNN; Inception	Yes	790/NR	Retrospective cohort; secondary analysis of data from Samsung Medical Center (Seoul, Korea)	2015	No
Choi et al (2018) ³⁸	Contrast-enhanced CT	NR	No	CNN	CNN	No	7461/NR	Three data cohorts: Cohort 1: retrospective cohort; secondary analysis of data from Asan Medical Center (Seoul, South Korea) used for development dataset Cohort 2: retrospective cohort; data collated from Inje University Paik Hospital and Hanyang University Hospital (both Seoul, South Korea) for external validation Cohort 3: retrospective cohort; data from Yonsei University Severance Hospital (Seoul, South Korea)	Cohort 1: 2007–16 Cohort 2: 2014–17 Cohort 3: 2010–11	No
Giropi et al (2017) ³⁹	CT	No	No	ConvNets	CNN	No	490 320/453	Retrospective cohort; secondary analysis of a subset of the multicentre Italian Lung Detection trial and the Danish Lung Cancer Screening Trial	NR	Yes
Codella et al (2017) ⁴⁰	Histology	NR	No	MA (ensembles)	NR	NR	900/NR	Retrospective cohort; data collected at multiple institutions for a research challenge (International Skin Imaging Collaboration)	NR	Yes
Coudray et al (2018) ⁴¹	Histology	No	No	Inception V3	CNN; Inception	Yes	825/148	Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of a subset from the Cancer Genome Atlas Program, National Cancer Institute Cohort 2: retrospective cohort; secondary analysis of data from the New York University Langone Medical Center (New York, NY, USA) for external validation	Cohort 1: NR Cohort 2: NR	Yes
De Fauw et al (2018) ⁴²	OCT	Yes	No	3D U-Net	CNN; U-Net	No	14 884/993	Retrospective cohort; secondary analysis of a subset of data collected at Moorfields Eye Hospital (London, UK)	2012–17	No

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Indicator definition			Algorithm		Data source		Open-access data			
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/tuning		Source of data	Data range	
(Continued from previous page)										
Ding et al (2019) ⁴³	PET	NR	Yes	Inception V3	CNN; Inception	Yes	1921/NR	Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of a prospectively collected dataset across the Alzheimer's Disease Neuroimaging Initiative Cohort 2: retrospective cohort; secondary analysis of data from the Department of Radiology, University of California (CA, USA) for external validation	Cohort 1: 2005–17 Cohort 2: 2006–16	Cohort 1: yes Cohort 2: no
Dunmon et al (2019) ⁴⁴	Chest x-ray	NR	Yes	ResNet-18; Dense-Net121; K SVM + BOVW; AlexNet	CNN; Residual Network; DenseNet Support Vector Machine	Yes	180 000/20 000	Retrospective cohort; secondary analysis of data from the Department of Radiology, Stanford University (CA, USA)	1998–2012	No
Elhteshami Bejnordi et al (2017) ⁴⁵	Histology	No	Yes	GoogLeNet; ResNet; VGG-16; VGG-Net; SegNet; 4-layer CNN; AlexNet; U-Net; CRFasRnn; 7-layer CNN, Agg-Net; 3-layer CNN	CNN; Inception; VGG; Residual Network; AlexNet; U-Net	Yes	270/NR	Retrospective cohort; data collected at the Radboud University Medical Center (Nijmegen, Netherlands) and University Medical Center (Urecht, Netherlands) for a research challenge ("CAMELYON16")	2015	Yes
Esteva et al (2017) ⁴⁶	Photographs	Yes	Yes	Inception V3	CNN; Inception	Yes	129 450/NR	Retrospective cohort; includes data from online open-access repositories (ie, the International Skin Imaging Collaboration Dermoscopic Archive, the Edinburgh Dermofit Library) and data from Stanford Hospital (CA, USA)	NR	Yes
Fujioka et al (2019) ⁴⁷	Breast ultrasound	NR	No	Inception V2	CNN; Inception	Yes	947/NR	Retrospective cohort; secondary analysis of data from Tokyo Medical and Dental University (Tokyo, Japan)	2010–17	No
Fujisawa et al (2019) ⁴⁸	Photographs	NR	No	GoogLeNet	CNN; Inception	Yes	4867/NR	Retrospective cohort; secondary analysis of data from University of Tsukuba Hospital (Tsukuba, Japan)	2003–16	No
Gómez-Valverde et al (2019) ⁴⁹	Fundus images	Yes	No	VGG-19	CNN; VGG	Yes	1560/NR	Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of data collated into the RIM-ONE and DRISHIT-GS datasets Cohort 2: retrospective cohort; secondary analysis of data from the Hospital de la Esperanza (Parc de Salut Mar; Barcelona, Spain)	Cohort 1: NR Cohort 2: NR	Cohort 1: yes Cohort 2: no
Grewal et al (2018) ⁵⁰	CT	NR	No	DenseNet	CNN; DenseNet	Yes	185/67	Retrospective dataset from two local hospitals	NR	No

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Indicator definition			Algorithm		Data source		Data range	Open-access data
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture	Transfer learning applied	Number of images for training/tuning	Source of data		
(Continued from previous page)								
Haenssle et al (2018) ³¹	NR	No	Inception V4	CNN; Inception	Yes	NR/NR	Cohort 1: NR Cohort 2: NR	Cohort 1: no Cohort 2: yes
					Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of data from the validated image library at the Department of Dermatology, University of Heidelberg (Heidelberg, Germany) Cohort 2: retrospective cohort; secondary analysis of a subset of data from the International Skin Imaging Collaboration melanoma project			
Hamm et al (2019) ³²	Yes	No	CNN	CNN	No	434/NR	2010–17	No
					Retrospective cohort; secondary analysis of data collected at the Department of Radiology and Biomedical Imaging, Yale School of Medicine (New Haven, CT, USA)			
Han et al (2018) ³³	No	Yes	Ensemble: ResNet-152 + VGG-19 (arithmetic mean of both outputs)	Ensemble, CNN; Residual Network	Yes	19 398/NR	2003–16	Yes
					Retrospective cohort; data collected at the Asan Medical Center, Inje, and Seoul University (Seoul, South Korea) and Hallym (Dongtan, South Korea)			
Han et al (2018) ³⁴	Yes	No	ResNet-152	CNN; Residual Network	Yes	49 567/3741	NR	Yes
					Retrospective cohort; data collected at the Asan Medical Center (Seoul, South Korea), University Medical Center Groningen (Groningen, Netherlands), Dongtan Sacred Heart Hospital (Gyeonggi, South Korea), Hallym University (Dongtan, South Korea), Sanggye Paik Hospital (Seoul, South Korea), Inje University (Seoul, South Korea), with additional data collected from open-access repositories and websites			
					Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of data collected from the imaging database of Seoul National University Hospital (Seoul, South Korea) Cohort 2: retrospective cohort; secondary analysis of data collected at Seoul National University Hospital (Seoul, South Korea), Boramae Medical Center (Seoul, South Korea), Kyunghee University Hospital at Gangdong (Seoul, South Korea), and Daejeon Eulji Medical Center (Daejeon, South Korea) Cohort 3: retrospective cohort; secondary analysis of the tuberculosis screening programme of Montgomery County (MD, USA) Cohort 4: retrospective cohort; secondary analysis of data from the tuberculosis screening programme of Shenzhen, China			
Hwang et al (2018) ³⁵	NR	Yes	CNN	CNN	No	60 689/450	Cohort 1: 2010–16 Cohort 2: 2016–17 Cohort 3: NR Cohort 4: NR	Cohort 1: no Cohort 2: no Cohort 3: yes Cohort 4: yes

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Indicator definition			Algorithm		Data source			Open-access data
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/ tuning)	Source of data	
(Continued from previous page)								
Hwang et al (2019) ⁵⁶	Yes	Yes	VGG-16; Inception V3; ResNet-50	CNN; VGG; Inception, Residual Network	Yes	28 720/7810	Retrospective cohort; secondary analysis of data collected at the Department of Ophthalmology of Taipei Veterans General Hospital (Taipei, Taiwan)	No
Hwang et al (2019) ⁵⁷	No	Yes	CNN	CNN	No	87 695/1050	Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of data from a single institution (no further details available) Cohort 2: retrospective cohort; data collated from four hospitals in Korea and four hospitals in France for external validation	No Cohort 1: 2010–17 Cohort 2: NR
Kernany et al (2018) ⁵⁸	Yes	Yes	Inception V3	CNN; Inception	Yes	OCT: 108 312/1000 Chest x-ray: 5232/NR	OCT: retrospective cohort of adult patients from the Shiley Eye Institute of the University of California San Diego (CA, USA), the California Retinal Research Foundation (CA, USA), Medical Center Ophthalmology Associates, Shanghai First People's Hospital (Shanghai, China), and Beijing Tongren Eye Center (Beijing, China) Chest x-ray: retrospective cohort of paediatric patients aged 1–5 years from Guangzhou Women and Children's Medical Center (Guangzhou, China)	Yes OCT: 2013, 2017 Chest x-ray: NR
Kim et al (2012) ⁵⁹	No	No	3-layer back propagation ANN (multilayered perceptron)	ANN	No	70/70	Retrospective cohort; data collected at the Kangwon National University College of Medicine (Gangwon-do, South Korea)	No
Kim et al (2018) ⁶⁰	No	No	CNN	CNN	No	NR/NR	Retrospective cohort; secondary analysis of data collected from Gangnam Severance Hospital (Seoul, South Korea)	No
Kim et al (2019) ⁶¹	No	Yes	ResidualNet	CNN; Residual Network	Yes	8000/1000	Retrospective cohort; secondary analysis of data collected from Seoul National University Hospital and Seoul National University Bundang Hospital (Seoul, South Korea)	No
Kise et al (2019) ⁶²	NR	No	AlexNet	CNN; AlexNet	Yes	400/NR	Retrospective cohort	No
Ko et al (2019) ⁶³	NR	No	CNN (imagenet-vgg-verydeep-16 and Imagenet-VGG-F)	CNN; VGG	Yes	594/NR	Retrospective cohort; secondary analysis of data collected at the Department of Radiology, Jeju National University Hospital, Jeju National School of Medicine (Jeju, South Korea)	No
Kumagai et al (2019) ⁶⁴	Yes	No	GoogLeNet	CNN; Inception	Yes	240/NR	Retrospective cohort; secondary analysis of data collected from Saitama Medical Center (Saitama, Japan)	No
Lee et al (2019) ⁶⁵	No	Yes	VGG-16; ResNet-50; Inception V3; Inception-ResNet-V2 ensemble	CNN; VGG; Residual Network; Inception	Yes	704/200	Retrospective cohort for development; both retrospective and prospective cohort for validation; data collected from Department of Radiology, Massachusetts General Hospital (Boston, MA, USA)	No

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Indicator definition			Algorithm		Data source		Data range	Open-access data	
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/ tuning			
(Continued from previous page)									
Li C et al (2018) ⁶⁶	Endoscopic photographs	Yes	Yes	CNN	CNN	Yes	5557/807	2008–17	No
Li X et al (2019) ⁶⁷	Ultrasound	Yes	Yes	ResNet-50 and Darnet-19, Ensemble DCNN	CNN; Residual Network	Yes	42 952/NR	2012–18	No
Lin et al (2014) ⁶⁸	Ultrasound	NR	No	Fuzzy Cerebellar Neural Network	Fuzzy Neural Network	No	NR/NR	2006–07	No
Lindsey et al (2018) ⁶⁹	X-ray	NR	Yes	DCNN	U-Net; DCNN	No	100 855/28 341	2016	No
Long et al (2017) ⁷⁰	Ocular photographs	No	No	CC-Cruiser; DCNN	DCNN	No	886/NR	Cohort 1: NR Cohort 2: 2012–16 Cohort 3: 2016	Cohort 1: no Cohort 2: no Cohort 3: yes
Lu W et al (2018) ⁷¹	OCT	Yes	Yes	ResNet	CNN; Residual Network	Yes	19 815/2202	2012–14	No
Matsuba et al (2019) ⁷²	Fundus images	NR	Yes	CNN	CNN	No	253/NR	NR	No
Nakagawa et al (2019) ⁷³	Endoscopic images	Yes	No	VGG	CNN; VGG	No	804/NR	2005–18	No
Nam et al (2019) ⁷⁴	Chest x-ray	Yes	Yes	DCNN	CNN	No	43 292/600	2010–17	No
Olczak et al (2017) ⁷⁵	Wrist, hand, and ankle x-rays	NR	No	VGG-16	CNN; VGG	Yes	179 521/NR	2002–15	No
Peng et al (2019) ⁷⁶	Fundus images	No	Yes	Pnet; LA-net; DeepSet Net; Inception V3; CNN; Dnet	CNN; Inception	Yes	58 402/NR	1992–2005	Yes
Poedjastoeiti et al (2018) ⁷⁷	X-ray	NR	Yes	VGG-16	CNN; VGG	Yes	400/NR	NR	No

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Indicator definition				Algorithm		Data source		Data range	Open-access data	
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/tuning)	Source of data			
(Continued from previous page)										
Rajpurkar et al (2018) ⁷⁸	Chest x-ray	NR	Yes	DenseNet	CNN; DenseNet	NR	98 637/6351	Retrospective cohort; secondary analysis of data within the ChestX-ray14 depository of the National Institutes of Health	NR	Yes
Raunviboonsuk et al (2019) ⁷⁹	Fundus images	Yes	No	CNN Inception-V4	CNN; Inception	NR	Not applicable/not applicable	Retrospective cohort; secondary analysis of data collected from the National Thai Registry of Diabetic Retinopathy	2015–2017	No
Sayres et al (2019) ⁸⁰	Fundus images	Yes	Yes	Inception V4	CNN; Inception	Yes	140 000/2000	Retrospective cohort; secondary analysis of data collected from Aravind Eye Hospital (India), Sankara Nethralaya Hospital (Chennai, India), and Narayana Nethralaya Hospital (India)	2015	No
Schlegl et al (2018) ²²	OCT	Yes	Yes	Deep learning model	CNN	No	840/NR	Retrospective cohort; secondary analysis of data collected from the Vienna Reading Center, Vienna, Austria	NR	No
Shibutani et al (2019) ⁸¹	SPECT	NR	Yes	ANN	ANN	No	Not applicable/not applicable	Prospective secondary analysis of data collected from Kanazawa University Hospital, Japan	NR	No
Shichijo et al (2017) ⁸²	Endoscopic images	Yes	No	GoogLeNet and Caffe DL framework	CNN; Inception	Yes	32 209/NR	Retrospective cohort; data collected at Tada Tomohiro Institute of Gastroenterology and Proctology, Saitama, Japan	2014–16	No
Singh et al (2018) ⁸³	Chest x-ray	No	Yes	"Qure AI"	CNN	No	1150 084/93 972	Two data cohorts: Cohort 1: retrospective cohort for training: "various hospitals" in India Cohort 2: retrospective cohort for testing: ChestX-ray8 database	Cohort 1: NR Cohort 2: NR	Cohort 1: no Cohort 2: yes
Song et al (2019) ⁸⁴	Ultrasound	NR	No	Multitask cascade pyramid CNN	CNN	Yes	6228/NR	Two data cohorts: Cohort 1: development data source not reported Cohort 2: retrospective cohort from open repository	Cohort 1: NR Cohort 2: NR	Cohort 1: no Cohort 2: yes
Stoffel et al (2018) ⁸⁵	Ultrasound	NR	No	Proprietary VIDI Suite deep-learning system	CNN	No	53/NR	Retrospective cohort; secondary analysis of data from the University Hospital Zurich, Switzerland	2013–15	No
Streba et al (2012) ⁸⁶	Contrast-enhanced ultrasound	NR	No	Multilayer ANN	ANN	No	NR/NR	Prospective cohort of patients that underwent contrast-enhanced ultrasound imaging, collected at the Research Center of Gastroenterology and Hepatology, Craiova, Romania	2008–11	No
Sun et al (2014) ⁸⁷	Transoesophageal echocardiogram	NR	No	ANN	ANN	No	NR/NR	Prospective cohort of patients collected at the Hospital of Harbin Medical University (Harbin, China)	2006–11	No
Tschandl et al (2019) ⁸⁸	Dermoscopy and clinical close-up	Yes	No	Inception V3	CNN; Inception	No	13 724/975	Two data cohorts: Cohort 1: retrospective cohort; secondary analysis of data from two open-source repositories (EDRA and ISIC) Cohort 2: retrospective cohort; secondary analysis of data from a single skin cancer centre	Cohort 1: NR Cohort 2: 2001–16	Cohort 1: yes Cohort 2: no

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Indicator definition			Algorithm		Data source			Open-access data
Method for predictor measurement	Exclusion of poor-quality imaging	Heatmap provided	Algorithm architecture name	Algorithm architecture	Transfer learning applied	Number of images for training/tuning	Source of data	
(Continued from previous page)								
Urakawa et al (2019) ⁸⁹	NR	No	VGG-16	CNN; VGG	Yes	2978/334	Retrospective cohort; secondary analysis of data collected from Department of Orthopedic Surgery, Tsuruoka, Municipal Shonai Hospital (Japan)	No
van Grinsven et al (2016) ⁹⁰	NR	Yes	SES CNN 60; NSES CNN 170	CNN	No	5287/NR	Retrospective cohort; secondary analysis of two open-access online repositories	Yes
Walsh et al (2018) ⁹¹	NR	No	Inception-ResNet-V2	CNN; Inception	Yes	929/89	Retrospective cohort; secondary analysis of data from La Fondazione Policlinico Universitario A Gemelli IRCCS (Rome, Italy) and University of Parma (Parma, Italy)	No
Wang et al (2017) ⁹²	NR	No	BP-ANN D13; BP-ANN T82; BP-ANN A95; BP-ANN S6; CNN	ANN	No	NR/NR	Retrospective cohort; data collected at the Affiliated Tumor Hospital of Harbin Medical University (Harbin, China)	No
Wang et al (2018) ⁹³	NR	No	3D CNN	CNN	No	1075/270	Retrospective cohort; secondary analysis of data collected from Fudan University Shanghai Cancer Center (Shanghai, China)	No
Wang et al (2019) ⁹⁴	NR	No	ResNet V2-50; YOLOv2	CNN; Residual Network	Yes	5007/NR	Retrospective cohort; secondary analysis of data from Affiliated Hospital of Qingdao University (China)	No
Wright et al (2014) ⁹⁵	Yes	No	CNN	CNN	No	257/NR	Retrospective cohort; data collected consecutively across two sites	No
Wu et al (2019) ⁹⁶	Yes	Yes	VGG-16; ResNet-50	CNN; VGG; Residual Network	Yes	24 549/NR	Retrospective cohort; secondary analysis of data from Renmin Hospital of Wuhan University (Wuhan, China)	No
Ye et al (2019) ⁹⁷	Yes	Yes	CNN; RNN	CNN; RNN	No	2255/282	Retrospective cohort; secondary analysis of data collected from three participating hospitals	No
Yu et al (2018) ⁹⁸	NR	No	CNN; VGG-16	CNN; VGG	Yes	362/109	Retrospective cohort; secondary analysis of data from Severance Hospital in the Yonsei University Health System (Seoul, South Korea) and Dongsan Hospital in the Keimyung University Health System (Daegu, South Korea)	No
Zhang C et al (2019) ⁹⁹	No	No	CNN	CNN	No	2285/757	Retrospective cohort; secondary analysis of data from Guangdong Provincial People's Hospital, Foshan First People's Hospital, and Guangdong Lung Cancer Institute (Guangdong, China) and The Third Affiliated Hospital of Sun Yat-Sen University and Guangzhou Chest Hospital (Guangzhou, China)	Yes
Zhang Y et al (2019) ¹⁰⁰	Yes	No	VGG-16	CNN; VGG	Yes	17 801/NR	Retrospective cohort; secondary analysis of data from Shenzhen Eye Hospital regional screening data (Shenzhen, China)	No
Zhao et al (2018) ¹⁰¹	NR	No	Denseshap; CNN	CNN; DenseNet	No	523/NR	Retrospective cohort; secondary analysis of data from Huadong Hospital Affiliated to Fudan University (Shanghai, China)	No
Number and type of predictors was not reported in any of the studies. NR=not reported. CNN=convolutional neural network. ANN=artificial neural network. RNN=recurrent neural network. DCNN=deep CNN. OCT=optical coherence tomography.								

Table 3: Indicator, algorithm, and data source for the 82 included studies

Number and type of predictors was not reported in any of the studies. NR=not reported. CNN=convolutional neural network. ANN=artificial neural network. RNN=recurrent neural network. DCNN=deep CNN. OCT=optical coherence tomography.

Table 3: Indicator, algorithm, and data source for the 82 included studies

and one study each for cardiology, gastroenterology or hepatology, gastroenterological or hepatological cancer, maxillofacial surgery, thyroid cancer, neurology, and nasopharyngeal cancer. These studies included 141 patient cohorts. Six studies included prospectively collected data, whereas all others used retrospective data. Nine studies used data from open-access repositories. In total, 161 contingency tables were included in the meta-analysis (appendix pp 3–6).

Hierarchical summary ROC curves of these 25 studies (161 contingency tables) are shown in figure 2. When averaging across studies, the pooled sensitivity was 88·6% (95% CI 85·7–90·9) for all deep learning algorithms and 79·4% (74·9–83·2) for all health-care professionals. The pooled specificity was 93·9% (92·2–95·3) for deep learning algorithms and 88·1% (82·8–91·9) for health-care professionals.

Of these 25 studies, only 14 used the same sample for the out-of-sample validation to compare performance between deep learning algorithms and health-care professionals, with 31 contingency tables for deep learning algorithm performance and 54 tables for health-care professionals (figure 3). The pooled sensitivity was 85·7% (95% CI 78·6–90·7) for deep learning algorithms and 79·4% (74·9–83·2) for health-care professionals. The pooled specificity was 93·5% (89·5–96·1) for deep learning algorithms and 87·5% (81·8–91·6) for health-care professionals.

After selecting the contingency table reporting the highest accuracy for each of these 14 studies (ie, 14 tables for deep learning algorithms and 14 tables for health-care professionals), the pooled sensitivity was 87·0% (95% CI 83·0–90·2) for deep learning algorithms and 86·4% (79·9–91·0) for health-care professionals. The pooled specificity was 92·5% (85·1–96·4) for deep learning algorithms and 90·5% (80·6–95·7) for health-care professionals (figure 4).

As an exploratory analysis, we also pooled performances of health-care professional and deep learning algorithms derived from matched internally validated samples (37 studies). Again, we selected a single contingency table for each study reporting the highest accuracy. In this sample, all accuracy metrics were higher, with a pooled sensitivity of 90·1% (95% CI 86·9–92·6) for deep learning algorithms and 90·5% (86·3–93·5) for health-care professionals and a pooled specificity of 93·3% (90·1–95·6) for deep learning algorithms and 91·9% (87·8–94·7) for health-care professionals (figure 4).

Discussion

To our knowledge, this is the first systematic review and meta-analysis on the diagnostic accuracy of health-care professionals versus deep learning algorithms using medical imaging. After careful selection of studies with transparent reporting of diagnostic performance and validation of the algorithm in an out-of-sample population, we found deep learning algorithms to have equivalent

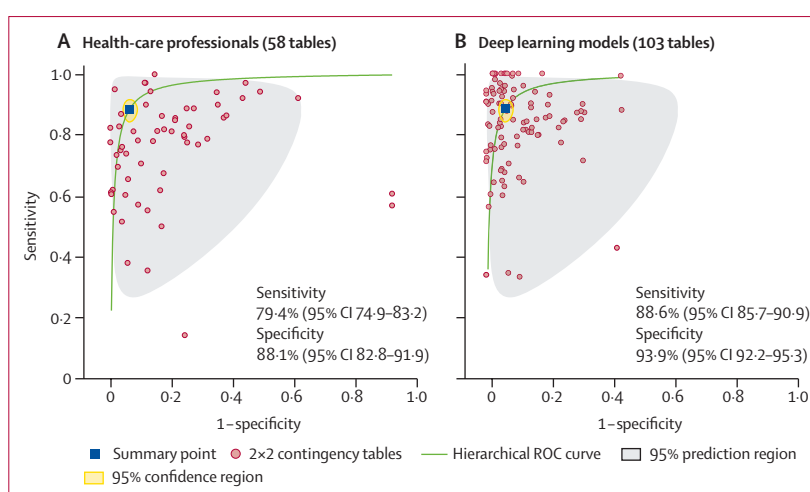


Figure 2: Hierarchical ROC curves of all studies included in the meta-analysis (25 studies)
ROC=receiver operating characteristic.

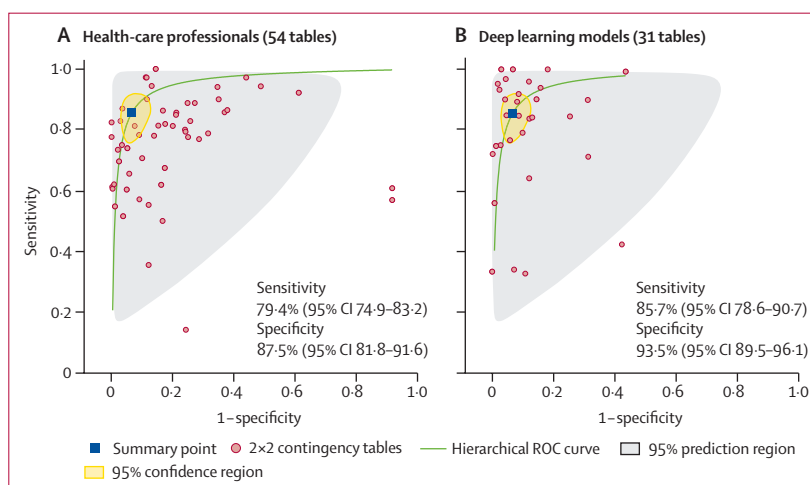


Figure 3: Hierarchical ROC curves of studies using the same out-of-sample validation sample for comparing performance between health-care professionals and deep learning algorithms (14 studies)
ROC=receiver operating characteristic.

sensitivity and specificity to health-care professionals. Although this estimate seems to support the claim that deep learning algorithms can match clinician-level accuracy, several methodological deficiencies that were common across most included studies should be considered.

First, most studies took the approach of assessing deep learning diagnostic accuracy in isolation, in a way that does not reflect clinical practice. Many studies were excluded at screening because they did not provide comparisons with health-care professionals (ie, human *vs* machine), and very few of the included studies reported comparisons with health-care professionals using the same test dataset. Considering deep learning algorithms in this isolated manner limits our ability to extrapolate the findings to health-care delivery, except perhaps for

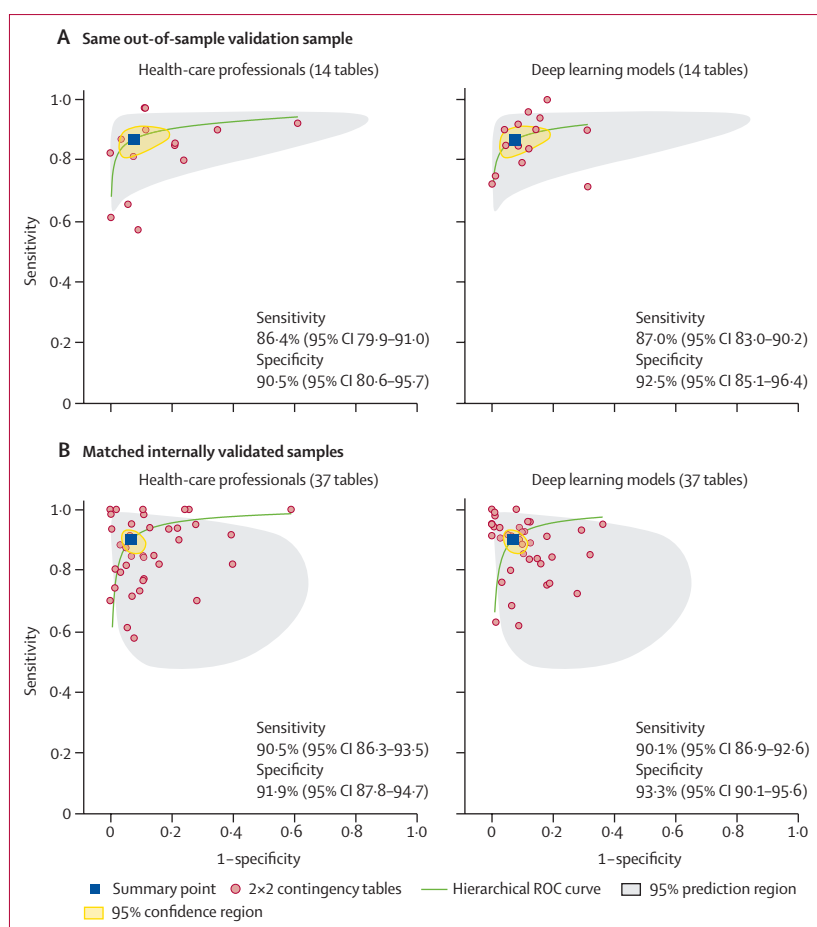


Figure 4: Hierarchical ROC curves of studies restricted to contingency tables reporting the highest accuracy Hierarchical ROC curves are shown for the 14 studies using the same out-of-sample validation sample to compare performance between deep learning algorithms and health-care professionals (A) and the 37 studies using matched internally validated samples (B). In both analyses, the single contingency table reporting the highest accuracy for each study was included. ROC=receiver operating characteristic.

mass screening.¹⁰² Only four studies provided health-care professionals with additional clinical information, as they would have in clinical practice; one study also tested the scenario in which prior or historical imaging was provided to the algorithm and the health-care professional, and four studies also considered diagnostic performance in an algorithm-plus-clinician scenario. It is worth noting that no studies reported a formal sample size calculation to ensure that the study was sufficiently sized in a head-to-head comparison. Although we acknowledge that sample size calculations can be challenging in this context, a lack of consensus on principled methods to perform them is no justification to ignore them in the design of a study.

Second, there were very few prospective studies done in real clinical environments. Most studies were retrospective, *in silico*, and based on previously assembled datasets. The ground truth labels were mostly derived from data collected for other purposes, such as in retrospectively collected routine clinical care notes or

radiology or histology reports, and the criteria for the presence or absence of disease were often poorly defined. The reporting around handling of missing information in these datasets was also poor across all studies. Most did not report whether any data were missing, what proportion this represented and how missing data were dealt with in the analysis. Such studies should be considered as hypothesis generating, with real accuracy defined in patients, not just datasets.

Third, a wide range of metrics were employed to report diagnostic performance in deep learning studies. If a probability function is not reported, the frequency of true positives, false positives, false negatives, and true negatives at a specified threshold should be the minimum requirement for such comparisons. In our review, only 12 studies reported the threshold at which sensitivity and specificity were reported, without justification of how the threshold was chosen; choice of threshold is often set at the arbitrary value of 0.5, as is convention in machine learning development. Metrics commonly used in the field of computer science, such as accuracy, precision, dice coefficient, and F1 score, are sometimes the only measure for reporting diagnostic performance. Since these tests are usually performed at a prevalence of 50%, these parameters are less comprehensive and useful for clinical practice.

Fourth, there is inconsistency over key terminology used in deep learning studies. Distinct datasets with independent samples should be defined in the development of a deep learning model from the initial training set through to one or more test sets that support validation. We found that the term “validation” is used variably, with some authors using the term appropriately for testing of the final model but others using it for the tuning of a model during development. It is crucial that the validation test set contains data independent to training or tuning data and is used only for assessing the final model. In several studies, we found a lack of transparency as to whether the test set was truly independent due to this inconsistent use of terminology. A standard nomenclature should be adopted. We suggest distinguishing the datasets involved in the development of an algorithm as training set (for training the algorithm), tuning set (for tuning hyperparameters), and validation test set (for estimating the performance of the algorithm). For describing the different types of validation test sets, we suggest adoption of the suggestion by Altman and Royston: internal validation (for in-sample validation), temporal validation (for in-sample validation with a temporal split), and external validation (for out-of sample validation).¹⁰³

Finally, although most studies did undertake an out-of-sample validation, most did not do this for both health-care professionals and deep learning algorithms. Moreover, only a small number of studies tested the performance of health-care professionals and deep learning algorithms in the same sample. In this review, we accepted both geographically and temporally split

test data, as well as the use of open-access datasets, as external validations. For internal validation, most studies adopted the approach of randomly splitting a single sample into training, tuning, and test sets, instead of preferred approaches such as resampling methods (eg, bootstrapping and cross validation), which have been recommended in clinical prediction model guidelines.¹⁸ Our finding when comparing performance on internal versus external validation was that, as expected, internal validation overestimates diagnostic accuracy in both health-care professionals and deep learning algorithms. This finding highlights the need for out-of-sample external validation in all predictive models.

An encouraging finding of this review is the improvement in quality of studies within the last year. 58 (71%) of the 82 studies satisfying the inclusion criteria were newly identified in the updated search, suggesting that the past year has seen a substantial increase in the number of studies comparing algorithm accuracy with health-care professionals. Only five studies additionally did external validation for algorithms and health-care professionals and were eligible for meta-analysis before the updated search, whereas a further 20 studies were suitable for meta-analysis in the review update. A persistent problem is studies not reporting contingency tables (or of sufficient detail for construction of contingency tables), as we were unable to construct contingency tables for two (9%) of 22 studies in the original search and 11 (18%) of 60 studies in the updated search.

Our final comparison estimating the differences in diagnostic accuracy performance between deep learning algorithms and health-care professionals is based on a relatively small number of studies. Less than a third of the included studies were eligible for meta-analysis. This is a direct consequence of poor reporting and lack of external validation in many studies, which has resulted in inadequate data availability and thus exclusion from the meta-analysis. We acknowledge that inadequate reporting does not necessarily mean that the study itself was poorly designed and, equally, that poor study design does not necessarily mean that the deep learning algorithm is of poor quality. Accordingly, there is considerable uncertainty around the estimates of diagnostic performance provided in our exploratory meta-analysis and we must emphasise that reliable estimates of the level of performance can only be achieved through well designed and well executed studies that minimise bias and are thoroughly and transparently reported.

We have not provided a systematic quality assessment for transparency of reporting in this review. This decision was made because existing reporting guidelines for prediction models, such as the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) statement, are focused primarily on regression-based model approaches, and there is insufficient guidance on how to appropriately apply

its checklist items to machine learning prediction models. The issues we have identified regarding non-standardisation of reporting in deep learning research are increasingly becoming recognised as a barrier to robust evaluation of AI-based models. A step in the right direction was the Delphi process undertaken by Luo and colleagues¹⁰⁴ to generate guidelines for developing and reporting machine learning predictive models. However, these guidelines have not been widely adopted, nor are they currently mandated by journals. An initiative to develop a machine learning version of the TRIPOD statement (TRIPOD-ML) was announced in April, 2019.¹⁰⁵

Although most of the issues we have highlighted are avoidable with robust design and high-quality reporting, there are several challenges that arise in evaluating deep learning models that are specific to this field. The scale of data required for deep learning is a well recognised challenge. What is perhaps less recognised is the way that this requirement skews the types of data sources used in AI studies, and the relative paucity of some of the associated data. For example, in many studies, historical registry data collected from routine clinical care or open-source databases are used to supply sufficient input data. These image repositories are rarely quality controlled for the images or their accompanying labels, rendering the deep learning model vulnerable to mistakes and unidentified biases. Population characteristics for these large datasets are often not available (either due to not being collected, or due to issues of accessibility), limiting the inferences that can be made regarding generalisability to other populations and introducing the possibility of bias towards particular demographics.

Traditionally, heavy emphasis for developing and validating predictive models is on reporting all covariates and model-building procedures, to ensure transparent and reproducible, clinically useful tools.¹⁰⁶ There are two main reasons why this is not possible in deep learning models in medical imaging. First, given the high dimensionality of the images, there are often too many individual datapoints driving predictions to identify specific covariates. Second, this level of influence and transparency of the algorithm is fundamentally incompatible with the black box nature of deep learning, where the algorithm's decisions cannot be inspected or explained. Few methods for seeing inside the black box—the black box deconvolution—are available, but new methods are being actively explored. An important example is the use of saliency or heat maps, which many studies adopt to provide some qualitative assessment of predictive features within the image.^{20,28,45,46,58,107} Other recent approaches such as influence functions and segmentation can offer additional information alongside saliency or heat maps.^{42,108} However, these approaches remain crude as they are limited to highlighting the location of salient features, rather than defining the pathological characteristics themselves, which would then allow a reproducible model to be built. Due to the

inability to interrogate a deep learning model, some caution should be exercised when making assumptions on a model's generalisability. For example, an algorithm could incorrectly form associations with confounding non-pathological features in an image (such as imaging device, acquisition protocol, or hospital label) simply due to differences in disease prevalence in relation to those parameters.^{109,110} Another consideration is the transparency of reporting deep learning model building procedures. These studies often do not report the full set of hyperparameters used, meaning the model cannot be reproduced by others. There are also issues of underlying infrastructure that pose similar challenges. For example, those building the AI model might use custom-built or expensive infrastructure that is simply not available to most research groups, and thus present concerns around reproducibility and the ability to scrutinise claims made in peer review. Cloud-based development environments can support code sharing between researchers without compromising proprietary information, but more work is needed to establish gold standards in reporting results in this domain.

Any diagnostic test should be evaluated in the context of its intended clinical pathway. This is especially important with algorithms where the model procedures and covariates cannot be presented explicitly. A randomised head-to-head comparison to an alternative diagnostic test, in the context of a clinical trial, could reveal and quantify possible clinical implications of implementing an algorithm in real life. Moreover, a common problem of test evaluation research could be overcome by testing these algorithms within a clinical trial: classification tasks are typically assessed in isolation of other clinical information that is commonly available in the diagnostic work-up.¹¹¹ Prospective evaluations of diagnostic tests as complex interventions would not only reveal the impact of these algorithms upon diagnostic yield but also on therapeutic yield.¹¹² In this context, the reporting of AI and machine learning interventional trials warrant additional consideration, such as how the algorithm is implemented and its downstream effects on the clinical pathway. In anticipation of prospective trials being the next step, extensions to the CONSORT and SPIRIT reporting guidelines for clinical trials involving AI interventions are under development.^{113–115}

Diagnosis of disease using deep learning algorithms holds enormous potential. From this exploratory meta-analysis, we cautiously state that the accuracy of deep learning algorithms is equivalent to health-care professionals, while acknowledging that more studies considering the integration of such algorithms in real-world settings are needed. The more important finding around methodology and reporting means the credibility and path to impact of such diagnostic algorithms might be undermined by an excessive claim from a poorly designed or inadequately reported study. In this review, we have highlighted key issues of design and reporting

that investigators should consider. These issues are pertinent for ensuring studies of deep learning diagnostics—or any other form of machine learning—are of sufficient quality to evaluate the performance of these algorithms in a way that can benefit patients and health systems in clinical practice.

Contributors

AKD, EJT, JRL, KB, LF, LMB, MKS, PAK, and XL contributed to the conception and design of the study. AK, AB, CK, DJF, GM, LF, MS, TM, SKW, and XL contributed to the literature search and data extraction. AKD, EJT, JRL, KB, LF, LMB, SKW, PAK, and XL contributed to data analysis and interpretation. AK, AB, CK, DJF, EJT, GM, MS, and SKW contributed to critical revision of the manuscript. AKD, JRL, LF, LMB, TM, SKW, PAK, and XL contributed to writing the manuscript, and all authors approved the manuscript. AKD, LMB, and PAK guarantee the integrity of the work. LF and XL contributed equally to this work.

Declaration of interests

PAK is an external consultant for DeepMind. JRL is an employee of DeepMind Technologies, a subsidiary of Alphabet. EJT has received personal fees from Verily and Voxel Cloud, and declare no support from any organization for the submitted work. LF, XL, AK, SKW, DJF, TM, AB, BM, MS, CK, MKS, KB, LMB and AKD have nothing to disclose.

Data sharing

The search strategy and extracted data contributing to the meta-analysis is available in the appendix; any additional data are available on request.

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